

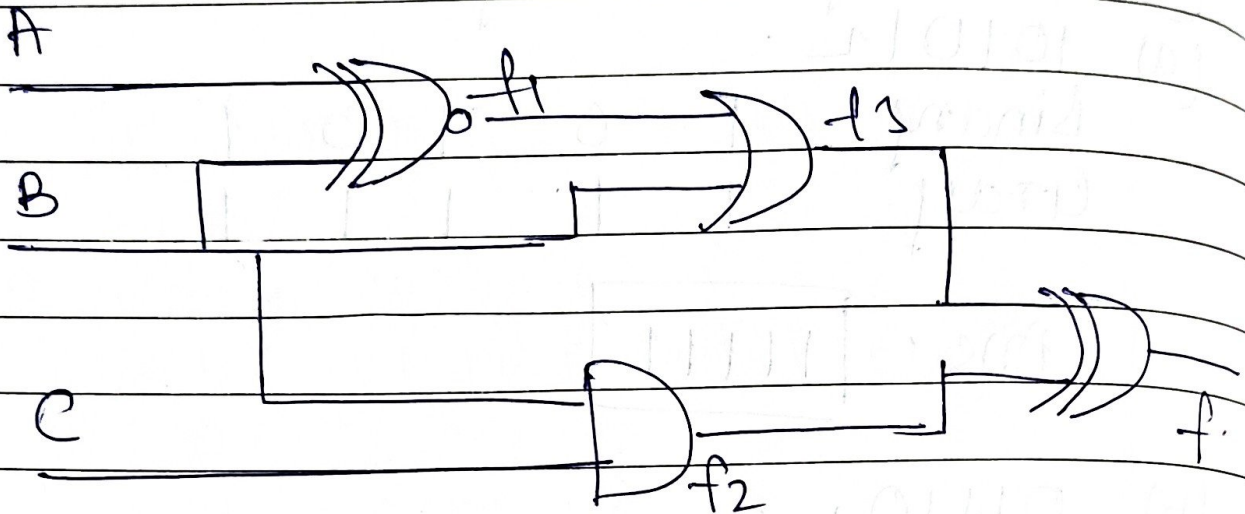
ADITI SHAILESH REVANKAR.

⑧

Assignment 2.

1.

(a)



$$f_1 = AB + A'B'$$

$$f_2 = BC$$

$$f_3 = (AB + A'B') + B$$

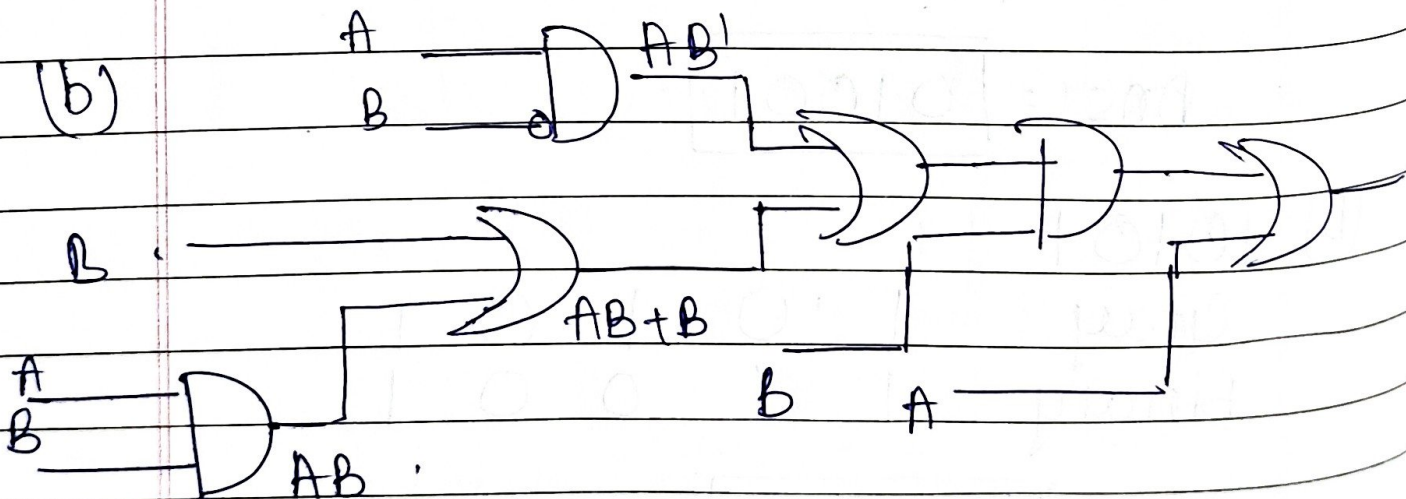
$$= (AB + B) = B$$

$$f_3 = A'B' + B + B = A'B' + B$$

$$f_3 = A' + B$$

$$f = (A' + B) \oplus BC$$

(b)



Q.

PAGE NO.

DATE

/ /

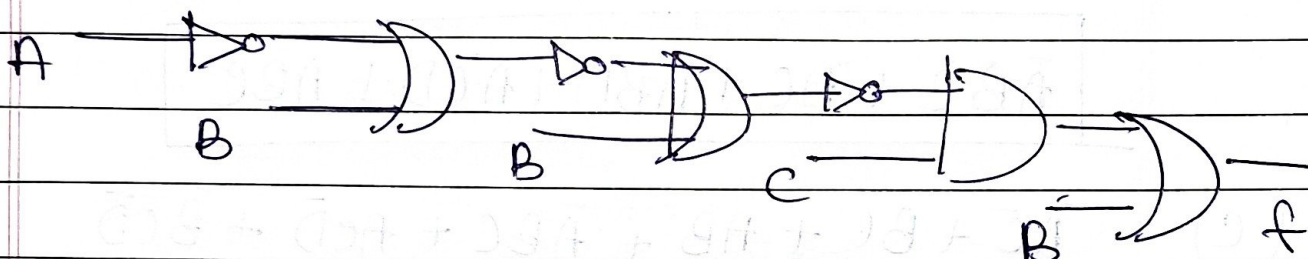
$$AB' + (AB + B)$$

$$AB' + (A) = A + B$$

$$(A + B)B = B$$

$$f = A + B$$

(c)



$$X = A' + B$$

$$X' = (A' + B)'$$

$$X' = AB'$$

$$Y = AB' \cdot B = 0$$

$$Y = 1$$

$$X = Z = 1, C = C$$

$$f = C + B$$

$$f = B + C$$



(10)

5 (a)  $ACDE + \bar{A}\bar{B}\bar{D} + ABCE + ABD$

The equation is already in minimal sop form.

(b)  $\bar{A}\bar{B}\bar{C} + ABD + \bar{A}C + \bar{A}C\bar{D} + A\bar{C}D + \bar{A}\bar{B}\bar{C}$

$$\bar{A}C + \bar{A}C\bar{D} = \bar{A}C$$

$$\bar{A}\bar{B}\bar{C} + \bar{A}C + ABD + A\bar{C}D + \bar{A}\bar{B}\bar{C}$$

(c)  $\bar{A}\bar{C} + BC + \bar{A}\bar{B} + \bar{A}BC + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$

$$BC + \bar{A}BC = BC$$

$$\bar{A}\bar{C} + BC + \bar{A}\bar{B} + A\bar{C}\bar{D} + \bar{B}\bar{C}\bar{D}$$

(d)  $\bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4 + \bar{x}_1x_2x_4 + x_1x_2\bar{x}_3x_4 + x_2$

$$x_2 + \bar{x}_1x_2x_4 + x_1x_2\bar{x}_3x_4 = x_2$$

$$x_2 + \bar{x}_1\bar{x}_3\bar{x}_4 + x_3x_4$$

11.

PAGE NO.

DATE / /

(3)(a)  $Z = A \oplus B \oplus C$

3 input XNOR gate

4(a)  $f(a,b,c,d) = \sum m(4,6,8,10,11,12,15)$   
 $+ \sum d(3,5,7,9)$

| cd \ ab          | ab | a $\bar{b}$ | $\bar{a}b$ | $\bar{a}\bar{b}$ |
|------------------|----|-------------|------------|------------------|
| $\bar{c}\bar{d}$ | 0  | 1           | X          | 2                |
| $\bar{c}d$       | X  | 4           | X          | 6                |
| $c\bar{d}$       | 8  | 12          | 13         | 14               |
| $cd$             | X  | 9           | X          | 10               |

$f = A\bar{B} + \bar{A}B + CD + \bar{C}\bar{D}$

(b)  $f(a,b,c,d) = \prod M(0,2,4,6,7,9) \cdot \prod D(10,11)$

| cd \ ab          | ab | a $\bar{b}$ | $\bar{a}b$ | $\bar{a}\bar{b}$ |
|------------------|----|-------------|------------|------------------|
| $\bar{c}\bar{d}$ | 0  | 1           | 3          | 2                |
| $\bar{c}d$       | 4  | 5           | 7          | 6                |
| $c\bar{d}$       | 8  | 12          | 13         | 14               |
| $cd$             | 9  | X           | X          | 10               |

$f = (A+D)(A+B+C)(\bar{A}+B+\bar{D})$